The Pollution Moat: Strategic Emissions under The Clean Air Act

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Abstract

Higher costs for entrants than incumbents under Clean Air Act nonattainment may induce incumbents to increase pollution to deter entry and strengthen market power. Consistent with this, following the 1997 PM2.5 standards, near-nonattainment-threshold firms increased emissions relative to those slightly farther away during the statutory pre-nonattainment monitoring window. The increase was stronger at PM2.5-intensive and near-monitor plants, in concentrated markets, and in high industry-growth-opportunities counties. Mechanisms include cross-plant emission reallocation within multi-plant firms and within-plant shifts in emission media toward air, with aggregate emissions unchanged. Following nonattainment, plant openings decline, and local emissions correlate positively with firm profits and valuations.

JEL Classification: Q58, Q52, L51, L13

Keywords: Clean Air Act, Air Pollution, Firm Behavior, Entry Barriers, Market Power

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